Scott Domains and Continuous Function Spaces

Small compact bases and an independent logical relation

Abstract

Scott-continuous maps between Scott domains with small compact bases form a Scott domain: directed and bounded suprema are pointwise, and its compact elements are exactly bounded finite joins of compact step functions. We give a constructive proof that keeps the smallness condition explicit and uses neither decidable order nor choice. We then use least fixed points and a tower of finite-stage projectors on the Scott filter domain to construct an independent logical relation. Its typed laws and fundamental lemma cover dependent products, annotated abstraction and function extensionality, and yield mere syntactic product inversion. An appendix develops the embedding–projection category, its equivalence with conditional upper semilattices, and initial algebras for endofunctors preserving countable chain colimits.

The function-space theorem is proved first because its compact basis and pointwise completeness underlie the later Scott filter construction. The fixed-point and projector-stage results then supply the semantic projector used in the logical relation. The appendix records the separate categorical theory of embedding–projection pairs.

1 Scott domains and small bases

We reason constructively. Fix a universe $\mathcal{U}$ of *small* types and a larger universe $\mathcal{U}^+$. Carriers of domains may belong to $\mathcal{U}^+$, whereas orders are small proposition-valued relations. Unless otherwise stated, all families whose suprema are considered have small index types. Existential assertions express mere existence: they do not include a chosen witness. These conventions are relevant to the proof; in a classical set-theoretic reading one may simply fix a size bound throughout.

Definition 1.1. Let D be a partially ordered set. An indexed family $(x_i)_{i\in I}$ is *directed* if I is inhabited and, for every $i, j \in I$, there exists $k \in I$ such that $x_i \le x_k$ and $x_j \le x_k$. It is *bounded* if there exists $u \in D$ such that $x_i \le u$ for every $i \in I$.

The order is *directed complete* if every small directed family has a supremum, and *bounded complete* if every small bounded family has a supremum. An element $a \in D$ is *compact* if, for every small directed family,

$$a \le \bigvee_{i\in I} x_i \quad\Longrightarrow\quad \exists i \in I\ a \le x_i.$$

Definition 1.2. A *Scott domain with a small compact basis* consists of a partially ordered set D, a least element $\bot_D$, directed and bounded completeness, and a small indexed family of compact elements

$$\beta_D : \mathsf{B}_D \longrightarrow D$$

such that, for every $x \in D$, the family

$$\mathsf{B}_D(x) = \{a \in \mathsf{B}_D \mid \beta_D(a) \le x\}$$

is directed and has supremum x.

For constructive and predicative domain theory with small bases, see de Jong and Escardó [3]. The basis need not be injectively indexed. We normally write a for $\beta_D(a)$ when an index $a \in \mathsf{B}_D$ is used as a domain element. The least-upper-bound part of the basis condition can equivalently be written

$$(\forall a \in \mathsf{B}_D\ (a \le x \implies a \le y)) \quad \implies \quad x \le y. \tag{1}$$

Our completeness convention is for small indexed families, not arbitrary families indexed by types in $\mathcal{U}^+$.

Lemma 1.3 (Finite common bounds)**.** *If finitely many compact elements $a_1, \ldots, a_n$ lie below the supremum of a directed family $(x_i)_{i \in I}$, then there exists $i \in I$ with $a_r \le x_i$ for every r. In particular, if $a_1, \ldots, a_n \le x$ in a Scott domain, there exists $a \in \mathsf{B}_D(x)$ above all the a_r. The statement includes $n = 0$.*

Proof. For each of the finitely many compact elements, compactness gives a family member above it. Directedness combines two such witnesses; finite induction combines all of them. For the empty family use inhabitedness of I. For the second assertion apply the first to the directed basis family with supremum x. Only finitely many existential witnesses are combined, so no choice principle is used. □

Lemma 1.4. *A bounded finite join of compact elements is compact. Every compact element of a Scott domain is equal to some basis element, in the sense of mere existence.*

Proof. Suppose $a = \bigvee_{r=1}^{n} a_r$ and $a \le \bigvee_i x_i$, where (x_i) is directed. Lemma 1.3 gives one x_i above every a_r, hence above a. This also proves compactness of the empty join $\bot$. If x is compact, apply its compactness to $x = \bigvee \mathsf{B}_D(x)$. Some $a \in \mathsf{B}_D(x)$ satisfies $x \le a$, and antisymmetry gives $x = a$. □

2 Continuous function spaces

The proof proceeds from pointwise completeness to compact steps, then to their bounded finite joins and a small basis.

2.1 Pointwise suprema

Definition 2.1. A function $f : D \to E$ is *Scott-continuous* if it is monotone and, for every small directed family (x_i),

$$f\left(\bigvee_i x_i\right) = \bigvee_i f(x_i).$$

Write $[D \Rightarrow E]$ for the set of these functions, ordered pointwise.

Lemma 2.2 (A small order on functions)**.** *For $f, g \in [D \Rightarrow E]$,*

$$f \le g \quad \iff \quad \forall a \in \mathsf{B}_D\ f(a) \le g(a).$$

In particular the order on $[D \Rightarrow E]$ can be expressed by a small proposition, even when the carrier of D is large.

Proof. Only the reverse implication needs proof. By continuity and the basis property, for every x,

$$f(x) = \bigvee_{a \in \mathsf{B}_D(x)} f(a) \le \bigvee_{a \in \mathsf{B}_D(x)} g(a) = g(x).$$

□

Proposition 2.3 (Pointwise suprema)**.** *The poset $[D \Rightarrow E]$ has a least element and all small directed and bounded suprema. Its least element is the constant function $\bot_E$, and these suprema are computed pointwise.*

Proof. Let $(f_i)_{i\in I}$ be a directed or bounded family of continuous maps. For every x, the corresponding family $(f_i(x))$ is directed or bounded, respectively, so define

$$h(x) = \bigvee_{i\in I} f_i(x).$$

Monotonicity of h follows from that of every f_i and the universal property of the supremum.

To prove continuity, let $(x_j)_{j\in J}$ be directed, with supremum x. Monotonicity already gives $h(x_j) \leq h(x)$. Suppose that z is any upper bound of all the $h(x_j)$. For every i, j we have $f_i(x_j) \leq h(x_j) \leq z$. Continuity of f_i gives $f_i(x) \leq z$, and therefore $h(x) \leq z$. Thus $h(x)$ is the supremum of the family $(h(x_j))$.

The pointwise universal property immediately makes h the supremum of the f_i in $[D \Rightarrow E]$. The constant bottom function is continuous and lies below every function. Taking the empty bounded supremum gives this same function. □

Lemma 2.4 (Finite input witnesses)**.** *If $f \in [D \Rightarrow E]$, $x \in D$, and b is compact in E, then*

$$b \leq f(x) \quad \Longrightarrow \quad \exists a \in \mathsf{B}_D(x)\ b \leq f(a).$$

Proof. Apply compactness of b to the directed supremum $f(x) = \bigvee_{a\in\mathsf{B}_D(x)} f(a)$. □

2.2 Compact step functions

Let a and b be compact elements of D and E, respectively. For each $x \in D$, the proposition $a \leq x$ is a small index type. The constant b family indexed by this proposition is bounded by b. We may therefore define

$$(a \Rightarrow_s b)(x) = \bigvee_{p:\, a\leq x} b. \tag{2}$$

This definition does not decide whether $a \leq x$. If $a \leq x$, its value is b; if $\neg(a \leq x)$, its value is $\bot_E$. These conditional statements do not assert a disjunction between the two cases. The subsingleton-supremum presentation of steps is also used by de Jong and Escardó [3].

Proposition 2.5. *The function $(a \Rightarrow_s b)$ is continuous and compact in $[D \Rightarrow E]$. For every continuous $f : D \to E$,*

$$(a \Rightarrow_s b) \leq f \quad \Longleftrightarrow \quad b \leq f(a). \tag{3}$$

Proof. Monotonicity follows by enlarging the indexing proposition. To check continuity on a directed family (x_i) with supremum x, let z bound all the values $(a \Rightarrow_s b)(x_i)$. If $a \leq x$, compactness of a gives an i with $a \leq x_i$, whence $b \leq (a \Rightarrow_s b)(x_i) \leq z$. The supremum property in (2) now gives $(a \Rightarrow_s b)(x) \leq z$. The opposite inequalities follow by monotonicity.

If $(a \Rightarrow_s b) \leq f$, evaluation at a gives $b \leq f(a)$. Conversely, if $b \leq f(a)$, then $a \leq x$ implies $b \leq f(x)$; the supremum property gives $(a \Rightarrow_s b)(x) \leq f(x)$.

Finally, suppose $(a \Rightarrow_s b) \leq \bigvee_i f_i$ for a directed family of continuous maps. By (3) and pointwise computation, $b \leq \bigvee_i f_i(a)$. Compactness of b gives an i with $b \leq f_i(a)$, and then $(a \Rightarrow_s b) \leq f_i$. □

2.3 Finite approximations

The function-space argument can be given directly using bounded completeness. Fix a continuous map $f : D \to E$. For compact basis elements $a_i \in \mathsf{B}_D$ and $b_i \in \mathsf{B}_E$ satisfying $b_i \leq f(a_i)$, put

$$h = \bigvee_{i=1}^{m} (a_i \Rightarrow_s b_i).$$

Each step lies below f by Proposition 2.5. Thus this finite family has the common upper bound f, and its supremum exists in the function space by pointwise bounded completeness. The empty join is the constant bottom function. There is no need to decide which entries are active at an input, or to construct a function by evaluating a graph.

Lemma 2.6. *Every bounded finite join of compact steps is compact. Moreover,*

$$\bigvee_{i=1}^{m} (a_i \Rightarrow_s b_i) \leq g \quad \Longleftrightarrow \quad \bigwedge_{i=1}^{m} b_i \leq g(a_i).$$

Proof. The characterization follows from the universal property of a supremum and the step-function characterization. For compactness, suppose the finite join lies below a directed supremum. Compactness of each step gives an index witnessing its inclusion in the directed family. Finitely many such witnesses have a common upper index, which then bounds their join. The empty case uses inhabitedness of the directed family. □

Definition 2.7. For $f : D \to E$ continuous, let $\mathcal{A}_f$ be the indexed family

$$\mathcal{A}_f = \left\{ \bigvee_{i=1}^{m} (a_i \Rightarrow_s b_i) \;\middle|\; a_i \in \mathsf{B}_D,\ b_i \in \mathsf{B}_E,\ b_i \leq f(a_i) \right\}.$$

The indices are finite lists of basis pairs with the displayed inequalities; in particular this is a small family.

Lemma 2.8. *The family $\mathcal{A}_f$ is directed and has supremum f.*

Proof. It contains the empty join. Combining two lists gives their join, still bounded by f, so it is directed.

Let h be its supremum. Clearly $h \leq f$. For the converse, fix $x \in D$ and a compact basis element $b \leq f(x)$. Continuity of f and algebraicity of D give a compact basis element $a \leq x$ with $b \leq f(a)$, by Lemma 2.4. The singleton step belongs to $\mathcal{A}_f$, and

$$b \leq (a \Rightarrow_s b)(x) \leq h(x).$$

Algebraicity of E gives $f(x) \leq h(x)$. □

2.4 The function-space theorem

Theorem 2.9. *If D and E are Scott domains, then $[D \Rightarrow E]$ is a Scott domain. Its compact elements are precisely the bounded finite joins of steps between compact basis elements.*

Proof. The pointwise constructions give bottom, directed completeness and bounded completeness. Lemmas 2.6 and 2.8 give algebraicity. If f is compact, apply its compactness to the directed family $\mathcal{A}_f$, whose supremum is f. Some member lies above f and also below f, hence equals f. Conversely, every member of every $\mathcal{A}_f$ is compact.

To obtain a single small compact basis, index finite lists of basis pairs satisfying the following small boundedness condition: every finite subfamily whose inputs have a common

basis upper bound has outputs with a common basis upper bound. Necessity follows by evaluating a bound of the corresponding steps at a common input bound. Conversely, for each x take the directed supremum of joins of output subfamilies whose inputs lie below x. Unions make this family directed, and its resulting function is the join of the steps by their universal property. Thus the lists satisfying the condition present all compact functions, and the families $\mathcal{A}_f$ establish density. □

Remark 2.10 (Local approximation and global smallness). The family $\mathcal{A}_f$ is small for each given f. The final part of the proof supplies one small basis for the entire function space, as required by Definition 1.2.

Remark 2.11 (Joint boundedness). For general Scott domains one must use joint boundedness of finite families. Pairwise compatibility alone is insufficient. Classically, the proper subsets of $\{1, 2, 3\}$ form a finite Scott domain under inclusion: the three singleton subsets are pairwise bounded but have no common upper bound. Taking a one-point input domain gives the corresponding counterexample to a pairwise test on steps.

3 Least fixed points

The following result requires only a pointed dcpo, although it applies in particular to every Scott domain and, by Theorem 2.9, to its continuous function spaces.

Theorem 3.1 (Kleene). *Let S be a pointed dcpo and $F : S \to S$ a Scott-continuous function. Define*

$$p_0 = \bot_S, \qquad p_{n+1} = F(p_n), \qquad p = \bigvee_{n\in\mathbb{N}} p_n.$$

Then $F(p) = p$. Moreover, for every $q \in S$,

$$F(q) \le q \quad \Longrightarrow \quad p \le q.$$

Thus $p = \mathrm{lfp}(F)$ is the least fixed point, and is also below every pre-fixed point of F.

Proof. Bottom gives $p_0 \le p_1$, and monotonicity of F gives inductively $p_n \le p_{n+1}$. Hence the sequence is directed. Continuity gives

$$F(p) = \bigvee_n F(p_n) = \bigvee_n p_{n+1} = \bigvee_n p_n = p,$$

where the last equality follows since the shifted chain is cofinal. If $F(q) \le q$, induction gives $p_0 \le q$ and $p_{n+1} = F(p_n) \le F(q) \le q$. Taking the supremum proves $p \le q$. □

In particular, after establishing continuity of a functional $\Phi : [D \Rightarrow E] \to [D \Rightarrow E]$, one can define its least solution by iteration from the constant bottom function. This construction assumes no idempotence, deflation or finitariness property of its iterates or of its limit. Any such properties must be proved separately.

Theorem 3.2 (Admissible fixed-point induction). *Let $F : S \to S$ be Scott-continuous on a pointed dcpo. Suppose a predicate $\mathcal{I}$ is closed under small directed suprema, contains $\bot$, and is preserved by F. Then*

$$\mathcal{I}(\mathrm{lfp}(F)).$$

No downward closure is required of $\mathcal{I}$.

Proof. Induction gives $\mathcal{I}(F^n(\bot))$ for every n; admissibility gives the conclusion at their supremum. This argument is proved once as a general rule. Its applications supply only admissibility, the bottom case, and preservation by the defining functional. □

`ScottAdmissible.seg` checks this rule for every presented Scott domain. The existing continuous binary-map interface has the same rule in `KleeneConstruction.seg`, using the directed-supremum closure interface of `BinaryAdmissible.seg`. Both rules allow invariant evidence in the larger universe; neither requires resizing it to a small proposition. They are derived theorems, not additional axioms.

4 Finite-stage projectors and projector compatibility

We now consider the Scott domain D of consistent filters generated by finite tokens. The tokens are built from $\bot$, U, products $\Pi(a, f)$, arrows $a \Rightarrow b$, and compatible intersections. Domain elements are ordered by inclusion of their token observations. Thus $\bot$ is least, and the intersection of compatible tokens represents their join of information. There are two different kinds of projector below. The binary semantic projector $p(a, x)$ enforces typing; the unary q_n observes a finite depth of the domain equation. Only the latter supplies the locality needed to construct the logical relation.

The filter domain comes with a fold–unfold isomorphism

$$\mathsf{unfold} : D \xrightarrow{\cong} \mathsf{H}(D), \qquad \mathsf{fold} : \mathsf{H}(D) \xrightarrow{\cong} D.$$

Here $\mathsf{H}(D)$ records a universe-head proposition, an optional product pair in $D \times [D \Rightarrow D]$, and a function component in $[D \Rightarrow D]$. The universe and product heads are disjoint, and either positive head forces the function component to bottom. For a Scott-continuous map $q : D \to D$ with $q(\bot) = \bot$, let $\mathsf{H}[q]$ retain the universe head, send a product pair (a, g) to $(q(a), q \circ g \circ q)$, and send the function component f to $q \circ f \circ q$. This is the projector action of the domain equation.

Proposition 4.1 (Finite-stage projector tower)**.** *Define q_0 by*

$$q_0(x) = \bigvee \{k \in \{\bot, \mathbf{u}\} \mid k \le x\},$$

where $\mathbf{u} \in D$ is the compact element represented by the universe token U. Define the remaining maps by the domain equation:

$$q_{n+1} = \mathsf{fold} \circ \mathsf{H}[q_n] \circ \mathsf{unfold}.$$

These are Scott-continuous deflations and satisfy

$$q_n^2 = q_n, \qquad q_n q_m = q_{\min(n,m)}, \qquad q_n \le q_{n+1} \le \mathrm{id}, \qquad \bigvee_n q_n = \mathrm{id}.$$

Put $D_n = \{x \in D \mid q_n(x) = x\}$, let $e_n : D_n \to D$ be inclusion, and let $r_n : D \to D_n$ send x to $q_n(x)$ with its fixed-point proof. Then

$$r_n e_n = \mathrm{id}_{D_n}, \qquad e_n r_n = q_n \le \mathrm{id}_D.$$

Thus $q_0(x)$ has no positive product or arrow observation; its only possible nonbottom compact observation is $\mathbf{u}$. Every compact element is fixed by some q_n. On a positive product observation, q_{n+1} sends its domain component through q_n and its family function g through $g \mapsto q_n \circ g \circ q_n$. It acts on an ordinary function component by the same function projection and preserves the universe observation. Consequently, components extracted from an element of D_{n+1} lie in, or factor through, D_n.

Proof. The indexing family defining $q_0(x)$ is inhabited by $\bot$, directed, and bounded by x; in particular $q_0(\bot) = \bot$. The head action preserves bottom, so every successive q_n also preserves bottom. For the successor, the displayed fold–unfold equation applies q_n to each domain component and $g \mapsto q_n \circ g \circ q_n$ to each function component. It defines a continuous map on filters independently of finite presentations.

Inductively, continuity, deflation and idempotence follow from the corresponding properties of q_n; for the function component use $q_n^2 = q_n$ and monotonicity of application. The same componentwise induction proves $q_n \leq q_{n+1}$ and the commuting equation. The stage-zero image and the asserted successor support are immediate from the defining clauses.

For exhaustivity, take a proper finite token presentation of a compact element and proceed by structural induction. The tokens $\bot$ and U are fixed by q_0. For an arrow token $a \Rightarrow b$, choose n such that the compact elements represented by a, b are fixed by q_n; the function-component equation makes the arrow fixed by q_{n+1}. For a product token $\Pi(a, f)$, the family token f is built from $\bot$, arrow tokens $a_i \Rightarrow b_i$, and finite intersections. Let F_f be the finite step function represented by f. Choose one n at which the compact elements represented by a and all a_i, b_i are fixed. Then $q_n \circ F_f \circ q_n = F_f$, so the product-component equation fixes $\Pi(a, f)$ at stage $n + 1$. Finally, if two compatible compact elements are fixed at stages m and n, their join is fixed at $\max(m, n)$: monotonicity puts both below the projection of their join, while deflation gives the reverse inequality. Thus every compact element is fixed by some q_n. Since these elements form a basis, $\bigvee_n q_n(x) = x$ for every x. □

These D_n are the fixed-image stages of the displayed fold–unfold construction on the filter-domain solution. Their projections have not been identified with those of the separate CUSL initial chain.

4.1 Finitary projections through compact semilattices

The general construction of Fp is particularly direct on the CUSL side of the Scott–CUSL equivalence (Theorem A.3). A CUSL is a partially ordered set with bottom and joins of compatible pairs. Its ideal completion is a Scott domain. The following construction uses its small presentation, without requiring decidable compatibility or a resizing axiom.

Theorem 4.2 (CUSL representation and the continuous Fp functor)**.** *Let C be a CUSL and $D = \mathrm{Idl}(C)$ its ideal completion. Finitary projections on D are represented by subsets $S \subseteq C$ containing bottom and closed under joins of pairs compatible in C:*

$$p_S(x) = \bigvee \{c \in S \mid c \leq x\}.$$

Such subsets are ordered by inclusion. They form an algebraic complete lattice whose compact elements are the finitely generated closed subsets. Their compact CUSL gives the CUSL presentation of $\mathrm{Fp}(D)$*.*

For a CUSL embedding $e : C \hookrightarrow C'$, direct and inverse image

$$e_*(S) = e[S], \qquad e^*(T) = e^{-1}[T]$$

form a continuous EP pair. This action satisfies the identity and composition laws and preserves every existing colimit of a small directed diagram of embeddings. Transport through the Scott–CUSL equivalence gives the continuous functor $D \mapsto \mathrm{Fp}(D)$ on Scott domains and EP pairs. For an EP pair $(e, r) : D \hookrightarrow E$, its embedding part is $P \mapsto ePr$.

Proof. For a closed subset S and an ideal x, the elements of S below x are inhabited, directed and bounded. Their downward closure is the ideal $p_S(x)$. This operation is continuous, deflationary and idempotent. Its compact fixed points are exactly S, and they give compact

fixed-point density. Conversely, a finitary projection is recovered from its compact fixed points. Thus the representation covers all finitary projector operations, not only a specified subclass. The subsets S need not be downward closed: a compact element below a fixed point need not itself be fixed.

Joins of closed subsets are obtained by closing their union under compatible joins; directed joins are already unions. Closure of a finite list of generators is compact. Every closed subset is the directed union of these finitely generated subsets, so they form a small compact basis. The checked compact-CUSL construction supplies smallness of the resulting object.

An embedding preserves and reflects compatibility and preserves bottom and compatible joins. Its direct images and inverse images therefore remain closed. They satisfy

$$e^* e_* = \mathrm{id}, \qquad e_* e^*(T) = T \cap e[C] \subseteq T.$$

Both operations preserve directed unions. Their identity and composition laws follow from those of images and inverse images.

For directed-colimit preservation, every element of the colimit CUSL comes from a stage. This coverage is derived from the categorical universal property: the union of the stage images inside the vertex is itself a CUSL colimit, and uniqueness identifies its inclusion with an isomorphism. Consequently every closed subset T of the vertex satisfies

$$T = \bigvee_i e_{i*} e_i^*(T).$$

If T is compact in the closed-family domain, it lies in one stage image. Its preimage there is compact because EP embeddings reflect compactness of elements in their images. Hence the mapped cocone also has stage coverage, which gives its full universal property. This proves preservation for arbitrary small directed index sets, not just countable chains. The checked transport of colimits through an equivalence gives the Scott-domain result. □

Corollary 4.3 (The finitary-projector adjunction)**.** *For every Scott domain D, inclusion $i_D : \mathrm{Fp}(D) \hookrightarrow [D \to D]$ has a continuous right adjoint κ_D, characterized by*

$$i_D(P) \leq f \quad \Longleftrightarrow \quad P \leq \kappa_D(f).$$

It satisfies $\kappa_D(i_D(P)) = P$ and $i_D(\kappa_D(f)) \leq f$; thus (i_D, κ_D) is an EP pair, as stated in CGW [1, p. 162]. In compact-family notation,

$$S_f = \{c \in K(D) \mid c \leq f(c)\}, \qquad \kappa_D(f)(x) = \bigvee \{c \in S_f \mid c \leq x\}.$$

The projection part of the functor in Theorem 4.2, for $(e, r) : D \hookrightarrow E$, is

$$Q \longmapsto \kappa_D(rQe).$$

Proof. The compact postfixed points S_f contain bottom and are closed under compatible joins. Their induced projector lies below f. Compact fixed-point density shows that every finitary projector below f lies below this one, proving the adjunction and retraction. A compact postfixed observation at a directed supremum of endomaps is already observed at one index. Hence κ_D preserves directed suprema. The inclusion does too, since directed unions of closed families give pointwise suprema of their projectors. The general-domain adjunction is obtained from the ideal-domain one by conjugating endomaps through the canonical ideal-completion isomorphism. Finally, inverse image of compact fixed-point families is exactly the family of compact postfixed points of rQe, proving the displayed restriction formula. □

These results are checked independently of the filter domain and its typing modules in `CUSLFpCheck.seg`. This session includes the CUSL construction, the general directed-colimit theorem, its transport to Scott domains, the adjunction and both action formulas. Source files and individual theorem names are listed in `UNIT/README.txt`.

4.2 The projector defined from the least fixed point

Define an operator on the Scott domain of continuous binary maps,

$$\Phi : [D \times D \Rightarrow D] \longrightarrow [D \times D \Rightarrow D].$$

For $Q : [D \times D \Rightarrow D]$, its value $\Phi(Q)$ is determined by the following equations on the positive heads of the domain equation:

$$\begin{aligned} \Phi(Q)(\mathbf{u}, \mathbf{u}) &= \mathbf{u}, \\ \Phi(Q)(\mathbf{u}, \pi(a, f)) &= \pi\big(Q(\mathbf{u}, a), z \mapsto Q(\mathbf{u}, f \cdot Q(a, z))\big), \\ \Phi(Q)(\pi(a, f), x) &= \big(z \mapsto Q(f \cdot Q(a, z), x \cdot Q(a, z))\big). \end{aligned}$$

Here $f \cdot y$ and $x \cdot y$ denote semantic application, and the last line denotes the corresponding function element of D. All other combinations of basic heads give $\bot$. For arbitrary elements, these equations are read through the constructive fold–unfold decomposition of D: each positive head is proposition-valued, and the three cases assemble into a continuous binary map without deciding which head is present. This construction is Scott-continuous in Q. Define p as the least fixed point of Φ. By the least-fixed-point theorem, it is computed by the Kleene sequence

$$P_0 = \bot, \qquad P_{k+1} = \Phi(P_k), \qquad p = \mathrm{lfp}(\Phi) = \bigvee_{k \in \mathbb{N}} P_k.$$

For any continuous binary operation Q, define its stage observation by

$$\rho_n(Q)(a, x) = r_n\big(Q(e_n a, e_n x)\big) \qquad (a, x \in D_n).$$

At this point p is a continuous binary operation satisfying $\Phi(p) = p$. Its projector laws follow from the closure theorem and invariant below, before comparison with inductive compact typing.

The dependent-product construction below follows Coquand, Gunter and Winskel [1, Proposition 8 and §7.2].

Theorem 4.4 (Dependent products of subdomains)**.** *Let A, D be Scott domains, and let $B : A \to \mathbf{D}^{\mathrm{EP}}$ be continuous, with coherent EP embeddings $B(a) \hookrightarrow D$. Then the continuous sections form a Scott domain*

$$\Gamma = \prod_{a \in A} B(a).$$

If A and $[D \to D]$ also embed in D as EP pairs, then Γ is a finitary subdomain of D. For literal subdomains $A = \mathrm{Fix}(P)$ and $B(a) = \mathrm{Fix}(Q_a)$, the intermediate EP pair is

$$E(s)(z) = s(Pz), \qquad R(g)(a) = Q_a(g(a)),$$

and its projector on $[D \to D]$ is

$$(ER)(g)(z) = Q_{Pz}(g(Pz)).$$

Proof. Represent a section by a continuous ambient-valued function $s : A \to D$ whose value at a is fixed by the fiber projection Q_a. The EP cone criterion converts continuity of B into continuity of these projections in the base argument. Conversely, a continuous family of finitary projections gives a continuous EP family of image domains. Both directions are checked using the explicit coherent inclusions and the directed supremum of embedding–projection composites.

Directed and bounded joins of sections are pointwise. A compact input $u \in A$ and a compact output $w \in D$ fixed by Q_u determine a compact section step. When it is active at a, monotonicity of the family makes w fixed by Q_a. Compatible finite joins of such steps form a small compact basis. To cover a section observation, continuity first gives a compact input witness, and finitariness of the corresponding fiber projection gives a compact fixed output witness. This proves algebraicity of Γ, without assuming an abstract section theorem. The recovered fiber components also satisfy the monotonicity and continuity equations for CGW sections.

The displayed E and R are continuous. Fiber fixedness and $P^2 = P$ give $RE = \mathrm{id}$. For $g : D \to D$,

$$Q_{Pz}(g(Pz)) \leq g(Pz) \leq g(z),$$

so $ER \leq \mathrm{id}$. Composing this EP pair with $[D \to D] \hookrightarrow D$ embeds the Scott domain Γ into D. Its induced projection is finitary: embeddings preserve compactness, so the compact basis of Γ gives ambient compact fixed-point density. The construction is checked for arbitrary presented Scott domains, separately from the filter model and all typing modules. □

Theorem 4.5 (Dependent sums and varying function spaces)**.** *Let A, C, D be Scott domains, let $a \mapsto Q_a$ be a continuous family of finitary projections on D, and let $A \hookrightarrow C$ be an EP pair (e, r). Then*

$$\Sigma = \sum_{a \in A} \mathrm{Fix}(Q_a)$$

is a Scott domain, ordered through its ambient components, with an EP embedding into $C \times D$. Its induced projection is

$$(c, z) \longmapsto (e(rc), Q_{rc}(z)).$$

If R is a finitary projection on D, the varying spaces

$$V(a) = [\mathrm{Fix}(Q_a) \to \mathrm{Fix}(R)]$$

have coherent embeddings into $[D \to D]$ and form a continuous EP family. Their total space is therefore a finitary subdomain of $C \times [D \to D]$, with projection

$$(c, g) \longmapsto (e(rc),\ z \mapsto R(g(Q_{rc}(z)))).$$

Proof. Represent a point of the first sum by (a, z) with $Q_a(z) = z$. Directed and bounded joins are inherited from $A \times D$. The compact basis consists of pairs (u, w) where u and w are ambient basis elements and $Q_u(w) = w$. To prove density, consider compact demands $v \leq a$ and $b \leq z$. Continuity of the family, applied to the basis approximation of a, gives a compact base support $u_0 \leq a$ with $b \leq Q_{u_0}(z)$. Choose a compact $u \leq a$ above v and u_0. Finitariness of Q_u then gives a compact w fixed by Q_u with $b \leq w \leq Q_u(z) \leq z$. Thus (u, w) meets both demands. This also gives directedness of the basis below each point. This is the coherent-subdomain form of the support argument in CGW [1, Proposition 7, pp. 132–134].

The embedding and projection are $E(a, z) = (e(a), z)$ and $P(c, z) = (rc, Q_{rc}(z))$. Joint continuity of $(a, z) \mapsto Q_a(z)$ proves continuity of P; $PE = \mathrm{id}$ and $EP \leq \mathrm{id}$ follow from the fixed-point condition and the given EP laws.

For the second assertion, the ordinary exponential EP action embeds $V(a)$ into $[D \to D]$. Its induced finitary projection is

$$T_a(g) = R \circ g \circ Q_a.$$

The exponential is order isomorphic to $\mathrm{Fix}(T_a)$. The family T_a is monotone and preserves directed suprema in a, by continuity of Q, g and R. Hence its fixed images form a continuous

EP family, and the first assertion applies. All these constructions, including the compact bases and EP laws, are checked for arbitrary presented Scott domains without filter or typing modules. Compactness is used within these general domain constructions; the application below uses their finitary projections. □

Remark 4.6 (Evaluation over varying domains). The same continuity mechanism appears in Coquand–Huber [2, p. 653 and Proposition 2, p. 658]. In their notation, consider

$$E = \sum_{a \in \mathrm{Type}} (\mathrm{El}(a) \times [\mathrm{El}(a) \to \mathrm{Type}]), \qquad \mathrm{ev}(a, v, f) = f(v).$$

The evaluation map is continuous. Consequently, for compact $w_0 \leq f(v)$ there is a compact triple $(a_0, v_0, f_0) \leq (a, v, f)$ in this total space such that

$$w_0 \leq f_0(v_0), \qquad v_0 \in \mathrm{El}(a_0), \qquad f_0 \in [\mathrm{El}(a_0) \to \mathrm{Type}].$$

Here comparisons between different fibers use their coherent ambient embeddings. The crucial point is that the compact argument and compact function have the *same* compact base code a_0. Proposition 2 uses this simultaneous approximation to reduce an observation of the codomain $f(v)$ to a finite product-type observation.

Theorem 4.5 supplies the corresponding varying function-space family and total-space support construction. Adding the argument component gives the displayed space E. Joint continuity of evaluation uses the same directed-diagonal argument as the continuous projectors above: for a directed family of ambient representatives,

$$(\bigvee_i f_i)(\bigvee_i v_i) = \bigvee_{i,j} f_i(v_j) = \bigvee_k f_k(v_k).$$

The last equality follows by choosing a common upper index for i, j. Algebraicity of the total space and compactness of w_0 then give the simultaneous witness. Thus the shared compact-base support principle also connects these domain constructions to the logical relation. The present Unit development checks the varying function spaces, dependent-sum support construction and joint-continuity arguments used above; the map ev on E and its compact-witness corollary are not yet packaged as separate checked theorems.

Theorem 4.7 (Normalized finitary families)**.** *Suppose $Q : D \times D \to D$ is continuous, each $Q(a, -)$ is a finitary projection, and*

$$Q(Q(\mathbf{u}, a), x) = Q(a, x).$$

Then $\Phi(Q)$ has the same properties. They hold for every Kleene iterate P_k and for p. In particular,

$$p(p(\mathbf{u}, a), x) = p(a, x).$$

Proof. Apply admissible fixed-point induction, Theorem 3.2, to the combined invariant

$$\mathcal{I}(Q) \quad \Longleftrightarrow \quad (\forall a,\ Q(a, -) \text{ is a finitary projection}) \ \wedge \ (\forall a, x,\ Q(Q(\mathbf{u}, a), x) = Q(a, x)).$$

The induction hypothesis concerns the whole continuous family, so it is available at every parameter occurring in a recursive call.

Preservation. Put $R = Q(a, -)$ and $S_v = Q(f(v), -)$. The function component is

$$T(g)(z) = S_{Rz}(g(Rz)).$$

Apply Theorem 4.4 to the Scott domain $\mathrm{Fix}(R)$ and the continuous EP family $v \mapsto \mathrm{Fix}(S_v)$. The resulting subdomain projector, after abstraction into D, is exactly the displayed function

component. The equality is checked in the filter instance of the general theorem. This proves finitariness of the function component without normalization.

For the universe/product component put $U_Q = \mathrm{Fix}(Q(\mathbf{u}, -))$ and $D_a = \mathrm{Fix}(Q(a, -))$ for $a \in U_Q$. Continuity of Q gives a continuous family $a \mapsto D_a$. By Theorem 4.5, the total space

$$\sum_{a \in U_Q} [D_a \to U_Q]$$

is a finitary subdomain of $D \times [D \to D]$. Its induced projection is

$$(a, f) \longmapsto \big(Q(\mathbf{u}, a),\ z \mapsto Q(\mathbf{u}, f(Q(Q(\mathbf{u}, a), z)))\big).$$

Normalization identifies $Q(Q(\mathbf{u}, a), z)$ with $Q(a, z)$, so this is precisely the universe/product payload operation of Φ. The equality is checked after abstraction in the filter model. Its compact fixed-point density supplies the code and function components simultaneously. Representing a compact function component by a finite arrow code and applying the product constructor gives a compact product code fixed by $\Phi(Q)(a, -)$. The clause proof thus uses the general total-space theorem; it no longer recursively assembles a common code from the entries of a finite graph. The universe head itself is fixed; absent heads contribute bottom. These cases give compact fixed-point coverage for $\Phi(Q)(a, -)$, hence idempotence and finitariness.

For normalization, deflation gives one inequality. The reverse follows by the observable-head decomposition. The product-type case uses $Q(Q(\mathbf{u}, a), -) = Q(a, -)$ and idempotence of both $Q(a, -)$ and $Q(\mathbf{u}, -)$; these identify its projected input and family in the function clause. The universe and bottom cases follow directly from their heads.

Bottom. The constant-bottom family satisfies both conditions.

Admissibility. Let (Q_i) be a small directed family satisfying $\mathcal{I}$, and put $Q_\infty = \bigvee_i Q_i$. A compact fixed point of an approximant remains fixed at the supremum, so compact fixed-point density passes to the supremum. Deflation passes pointwise, and compact fixed-point coverage gives idempotence. For normalization, continuity reduces the nontrivial inequality to mixed indices n, m. Choose $i \geq n, m$ and use

$$Q_n(Q_m(\mathbf{u}, a), x) \leq Q_i(Q_i(\mathbf{u}, a), x) = Q_i(a, x) \leq Q_\infty(a, x).$$

The reverse inequality follows from the diagonal terms. Thus $\mathcal{I}$ is admissible. The induction rule now gives $\mathcal{I}(p)$. This application contains no induction on the Kleene iterates: `IndependentKleeneProjector.seg` supplies the three obligations and invokes the rule. The independent projector checker session checks that application and its preservation and admissibility lemmas without the typed projector comparison. □

Lemma 4.8 (One unfolding lowers the comparison stage)**.** *For all continuous binary operations Q, R,*

$$\rho_0(\Phi(Q)) = \rho_0(\Phi(R)).$$

For every n,

$$\rho_n(Q) = \rho_n(R) \quad \Longrightarrow \quad \rho_{n+1}(\Phi(Q)) = \rho_{n+1}(\Phi(R)).$$

Proof. At stage zero only the universe head can be observed, besides bottom. The universe head of $\Phi(Q)(A, X)$ is present exactly when both A and X have a universe head; this does not depend on Q.

For the successor case, use the structural equations of Proposition 4.1. On a product q_{n+1} projects the domain with q_n and the family function by $g \mapsto q_n \circ g \circ q_n$. On the function component it uses the same function projection, and it preserves the universe head.

If A, X belong to stage $n + 1$, the domain component of either product head belongs to stage n. Each family function, and the application function of X, takes values in stage n and

factors through q_n. Consequently, in the nested calls occurring in $\Phi(Q)$, a function applied to $Q(e_n a, e_n z)$ depends only on

$$r_n Q(e_n a, e_n z) = \rho_n(Q)(a, z).$$

Its output is itself an embedded stage-n value. The outer recursive call, after projection by r_n, is therefore also determined by $\rho_n(Q)$.

For example, the restricted function clause is determined by

$$z \longmapsto \rho_n(Q)\big(B(\rho_n(Q)(a, z)),\ F(\rho_n(Q)(a, z))\big),$$

where a is the product's stage-n domain component, B its family, and F the stage-n application function of X. The product clause has the same calculation with the constant universe family. Replacing $\rho_n(Q)$ by $\rho_n(R)$ gives equal components and hence equal stage-$(n+1)$ results. These arguments use positive head observations; no decidability of a head is required. □

Corollary 4.9 (Finite stabilisation)**.** *For $k, j \geq n+1$,*

$$\rho_n(P_k) = \rho_n(P_j).$$

Proof. Induct on n. The base is the first assertion of Lemma 4.8. At the successor, write $k = k' + 1$ and $j = j' + 1$, apply the induction hypothesis to $P_{k'}, P_{j'}$, and then apply the second assertion of the lemma. □

Define the stage operation directly from the least fixed point:

$$p_n(a, x) = r_n\big(p(e_n a, e_n x)\big), \qquad a, x \in D_n.$$

Theorem 4.10 (Finite computation of the stage operation)**.** *For every $a, x \in D_n$,*

$$p_n(a, x) = r_n\big(P_{n+1}(e_n a, e_n x)\big).$$

Proof. Continuity of r_n and pointwise computation of the Kleene supremum give

$$\rho_n(p) = \bigvee_k \rho_n(P_k).$$

The increasing family on the right is constant from index $n+1$ onwards, by Corollary 4.9, so its supremum is $\rho_n(P_{n+1})$, which is the displayed right-hand side. □

Thus p_n can be computed using only $n+1$ unfoldings of Φ. This theorem does not prove that $p_n(a, -)$ is a finitary projection: finite stabilisation alone gives neither deflation nor idempotence. The checked stage finitariness proof used later identifies p_n with the projector generated by bounded inductive compact typing. Theorem 4.7 supplies independent global finitariness, but the stage conclusion requires an additional argument. Arbitrary EP conjugation rPe need not be idempotent: on the finite chain $0 < 1 < 2 < 3$, take $P = (0, 0, 2, 2)$, embed $\{0, 1, 3\}$, and project 2 to 1. Then rPe sends 3 to 1 and 1 to 0. The finite-order laws and this failure are checked in `EPRestrictionObstruction.seg`. Compatibility with the particular rank projections is therefore needed to remove the bounded-typing stage proof.

The continuous functor Fp and its adjunction are now checked in general (Theorem 4.2 and Corollary 4.3). Its projection part sends a projector P to $\kappa_{D_n}(r_n P e_n)$. This produces a finitary projector, but the direct stage formula above uses the raw restriction $r_n P e_n$. To obtain an independent proof of that formula's finitariness, one must show that these particular restrictions are unchanged by κ_{D_n}. The existing bounded-typing proof establishes stage finitariness; the general functor theorem does not replace the required rank compatibility argument.

The logical-relation construction and fundamental lemma below use only the projector-tower properties of Proposition 4.1.

5 An independent logical relation

5.1 Admissible predicates and the relation clauses

We now use the Scott filter domain to relate denotations to syntax. Write $\mathbf{u} \in D$ for the universe element, $\pi(a, \varphi)$ for the product constructor, and $f \cdot x$ for application. A family φ is a function component of D, representing a continuous map $D \to D$. The recursive domain equation supplies universe observations and product observations. A positive product observation

$$\pi(\bot, \bot) \leq x$$

allows extraction of a domain component x_{d} and a family component x_{f}. These components are only used under that positive hypothesis. No decision whether an arbitrary element has a head is needed. The product $\pi(\bot, \bot)$ is a positive compact observation, distinct from $\bot$.

Fix a syntactic context Δ. Terms have variables, a universe U, dependent products $\Pi(A, B)$, annotated abstractions $\lambda A.M$, and applications. The body of a product or abstraction has one additional variable; BN denotes substitution of N for it. More generally, $A\sigma$ is substitution by the simultaneous substitution σ. Write $T \longrightarrow^* V$ for head reduction and $\Delta \vdash M = N : A$ for typed definitional equality. When such a judgement occurs inside $\| \cdot \|$, it denotes the type of its derivations, and the brackets denote propositional truncation.

5.1.1 Lower admissible predicates

Definition 5.1. A small proposition-valued predicate Q on D is *lower admissible* if it is downward closed and closed under small directed suprema. It is *strict* if $Q(\bot)$ holds. Strictness is a separate condition: directed families are inhabited, so admissibility alone does not imply it.

Lemma 5.2 (Compact reconstruction)**.** *For every lower admissible predicate Q,*

$$Q(x) \quad \Longleftrightarrow \quad \forall k \in \mathsf{B}_D \; (k \leq x \Longrightarrow Q(k)).$$

Conversely, for a small predicate R on the compact basis,

$$\overline{R}(x) = \forall k \in \mathsf{B}_D \; (k \leq x \Longrightarrow R(k))$$

is lower admissible. If R respects the basis order, then $\overline{R}(k) \Longleftrightarrow R(k)$ for every basis element k.

Proof. Downward closure proves one direction of reconstruction; the directed basis of x and admissibility prove the other. For admissibility of $\overline{R}$, a compact observation below a directed supremum lies below one member. Its mere witness may be eliminated into the proposition $R(k)$. Recovery on a compact uses downward closure and the observation of that compact itself. □

For lower admissible predicates P, Q, define preservation by

$$\mathrm{Arr}(P, Q)(f) \quad \Longleftrightarrow \quad \forall x \in D \; (P(x) \Longrightarrow Q(f \cdot x)).$$

Although this formula quantifies over a potentially large domain, it is equivalent to the small formula

$$\forall k \in \mathsf{B}_D \; (P(k) \Longrightarrow Q(f \cdot k)). \tag{4}$$

Indeed, if $P(x)$, then $P(k)$ for every compact $k \leq x$; continuity of application and admissibility of Q give the conclusion at x. For each compact input, application is continuous in f. Thus (4) is an intersection of lower admissible predicates of f, and is itself lower admissible. It is contravariant in P and covariant in Q.

5.1.2 Type and term predicates

We seek predicates

$$\mathsf{Ty}(x; T, T'), \qquad \mathsf{Elem}(x; T, M, M')$$

for all syntactic indices in Δ. Each is to be small, strict and lower admissible in x. There is no separate semantic classifier argument in Elem: semantic typing will enter its laws and the fundamental lemma.

For candidate term predicates P and Q, define the function condition $\mathcal{F}(P, Q)(f; T, M, M')$ as follows. For every reduction $T \longrightarrow^* \Pi(A, B)$, every derivation $\Delta \vdash N = N' : A$, and every x with $P(x; A, N, N')$, require

$$\begin{aligned} &Q(f \cdot x; BN, MN, MN'), \\ &Q(f \cdot x; BN, M'N, M'N'), \\ &Q(f \cdot x; BN, MN, M'N'). \end{aligned} \tag{5}$$

All three comparisons use the same left result type BN. For candidate term predicates P and type predicates Q, the family condition $\mathcal{G}(P, Q)(\varphi; A, B, B')$ similarly requires

$$\begin{aligned} &Q(\varphi \cdot x; BN, BN'), \\ &Q(\varphi \cdot x; B'N, B'N'), \\ &Q(\varphi \cdot x; BN, B'N') \end{aligned} \tag{6}$$

whenever $\Delta \vdash N = N' : A$ and $P(x; A, N, N')$. The compact-input formulation makes both conditions small and lower admissible in their semantic argument.

The type condition $\mathcal{T}(P, Q)(x; T, T')$ has two parts. First, a universe observation $\mathbf{u} \leq x$ implies

$$\|T \longrightarrow^* U \ \times \ T' \longrightarrow^* U\| .$$

Second, a positive product observation implies the mere existence of A, A', B, B' with

$$\begin{aligned} &T \longrightarrow^* \Pi(A, B), \qquad T' \longrightarrow^* \Pi(A', B'), \\ &\Delta \vdash A = A' : U, \qquad \Delta, A \vdash B = B' : U, \\ &Q(x_{\mathrm{d}}; A, A'), \qquad \mathcal{G}(P, Q)(x_{\mathrm{f}}; A, B, B'). \end{aligned} \tag{7}$$

The whole tuple in (7) is truncated. In particular, its component equalities are retained even at the smallest positive product observation.

Lemma 5.3 (Admissibility of the type condition)**.** *The condition $\mathcal{T}(P, Q)$ is lower admissible whenever the candidate predicates are lower admissible. It is contravariant in P and covariant in Q. It is strict when the positive predicates are strict.*

Proof. Head determinism says that any two evidenced product reducts of the same pair T, T' have identical syntactic components. Consequently the product condition can equivalently be expressed by a mere existence of evidenced heads, together with the component conditions for every such evidenced head. This separates head existence from universal component tests.

Each head-existence requirement is an observation implication $k \leq x \Longrightarrow E$, for a fixed compact head k and a small proposition E. Such an implication is lower admissible by compactness. The domain component can be tested by observations $\pi(k, \bot) \leq x$; the family component by observations $\pi(\bot, k) \leq x$. Compact reconstruction identifies these tests with the required predicates on the extracted components. Intersections give admissibility of the whole condition. Variance follows from that of argument preservation. Bottom has neither a universe head nor a positive product head, and the remaining strict output conditions are inherited from Q. □

The desired invariant equations are

$$\begin{aligned} \mathsf{Ty}(x;T,T') &\iff \mathcal{T}(\mathsf{Elem},\mathsf{Ty})(x;T,T'),\\ \mathsf{Elem}(x;T,M,M') &\iff \mathsf{Ty}(x;M,M') \ \wedge\ \mathcal{F}(\mathsf{Elem},\mathsf{Elem})(x;T,M,M'). \end{aligned} \tag{8}$$

These equations are mixed in variance. In particular their simultaneous diagonal action is not a monotone operator to which Kleene iteration can simply be applied.

5.2 Construction by compatible finite depths

Use the projectors q_d constructed above. They satisfy $q_d \le q_{d+1} \le \mathrm{id}$, $q_d q_e = q_{\min(d,e)}$, and $\bigvee_d q_d = \mathrm{id}$. Every compact basis element is fixed by some q_d. Two pairs of predicates *agree at depth d* if their type and term components are equivalent on every element fixed by q_d, for all syntactic indices.

For a pair $R = (R^{\mathrm{ty}}, R^{\mathrm{tm}})$, form the new type predicates before the new term predicates:

$$\begin{aligned} (R^+)^{\mathrm{ty}}(x;T,T') &= \mathcal{T}(R^{\mathrm{tm}},R^{\mathrm{ty}})(x;T,T'),\\ (R^+)^{\mathrm{tm}}(x;T,M,M') &= (R^+)^{\mathrm{ty}}(x;M,M') \ \wedge\ \mathcal{F}(R^{\mathrm{tm}},R^{\mathrm{tm}})(x;T,M,M'). \end{aligned} \tag{9}$$

Reading the newly formed type component in the second line is essential. The original mutual action reads an old type component at the same depth and does not directly have the following one-step locality property.

Lemma 5.4 (Locality)**.** *If R and S agree at depth d, then R^+ and S^+ agree at depth $d+1$.*

Proof. For an element fixed by q_{d+1}, every extracted product domain is fixed by q_d. Its family function, and its application function, have depth-d input and output support. Write h for either such function. Then

$$h(x) = h(q_d x), \qquad q_d(h(x)) = h(x).$$

Suppose the arrow condition for R is known and an argument is related by the negative predicate of S. Lower its evidence to $q_d x$. Agreement at depth d converts this evidence to the negative predicate of R. Apply the given arrow condition at $q_d x$, use the first equation to identify the output with $h(x)$, and use the second equation to convert the positive output predicate from R to S.

This proves both directions for functions and families; the domain component is handled directly by agreement at depth d. Head evidence is unchanged. Thus the new type predicates agree at depth $d+1$. Using that agreement in the new term clause finishes the proof. □

Define the initial pair by the universe-observation tests

$$\begin{aligned} R_0^{\mathrm{ty}}(x;T,T') &= (\mathbf{u} \le x \Longrightarrow \|T \longrightarrow^* U \times T' \longrightarrow^* U\|),\\ R_0^{\mathrm{tm}}(x;T,M,M') &= R_0^{\mathrm{ty}}(x;M,M'), \qquad R_{d+1} = R_d^+. \end{aligned}$$

At depth zero, every element lies below $\mathbf{u}$, has no product head, and its applications are bottom. Hence R_0 agrees at depth zero with R^+ for every strict R. This statement does not require deciding whether a depth-zero element is bottom or the universe.

Lemma 5.5 (Compatibility and recovery)**.** *Each R_d is strict and lower admissible, and*

$$d \le e, \quad q_d x = x \quad \Longrightarrow \quad R_d(x) \iff R_e(x).$$

Here and below such formulas apply separately to each component and all of its syntactic indices. Define

$$R_\infty(x) = \forall d \in \mathbb{N}\ R_d(q_d x). \tag{10}$$

Then R_∞ is strict and lower admissible, and

$$q_d x = x \quad \Longrightarrow \quad R_\infty(x) \Longleftrightarrow R_d(x).$$

Proof. The depth-zero observation and Lemma 5.4 prove compatibility by induction. Strictness and admissibility are preserved at every step. The limit is a small countable intersection of continuous pullbacks of lower admissible predicates, so has the same properties.

For recovery, the forward implication uses the dth conjunct. Conversely suppose $q_d x = x$ and $R_d(x)$. If $d \le e$, compatibility gives $R_e(x)$, which lowers to $R_e(q_e x)$. If $e \le d$, first lower to $R_d(q_e x)$ and then use compatibility at the depth-e element $q_e x$. Constructive comparison of natural numbers gives every conjunct. □

Theorem 5.6 (Independent invariant)**.** *There is a unique strict lower admissible pair* $(\mathsf{Ty}, \mathsf{Elem})$ *satisfying* (8)*. It is given by* (10)*.*

Proof. Recovery and locality show that R_∞ and R^+_∞ agree at every finite depth. Every compact is captured at a finite depth, so they agree on all compact observations. Lemma 5.2 gives agreement on all domain elements. The type equation then identifies the fresh type component in (9) with R^{ty}_∞, yielding both original mutual equations.

For uniqueness, a strict solution of the original equations also solves the type-before-term equations. The depth-zero observation and locality show inductively that it agrees with R_d at depth d. Recovery and compact reconstruction identify it with R_∞. □

This is a finite-depth construction with an admissible limit, rather than iteration of a monotone diagonal relation operator. It constructs the relation without an earlier logical relation or adequacy theorem. All existential eliminations have proposition-valued targets. In particular, it never replaces $\forall i\ \|W_i\|$ by $\|\forall i\, W_i\|$.

5.3 Typed laws and restriction of arguments

Write $v : c$ for the inductive compact typing judgement. For $a \in D$, let $T(a)$ be the family generated by compact values v for which $v : c$ and $c \le a$, closed under compatible finite joins and compact equivalence. The associated map is

$$p_{\mathrm{ty}}(a, x) = \bigvee \{v \in T(a) \mid v \le x\}.$$

It is a finitary projection: it is continuous, lies below the identity, is idempotent, and every fixed point is a directed supremum of compact fixed points. The operation p_{ty} defined from $T(a)$ satisfies the three recursive equations for Φ and lies below every continuous operation satisfying them. Consequently $p_{\mathrm{ty}} = p$, the least fixed point defined in §4.1. This comparison is used for typed laws below. Finitariness and normalization of both p and all iterates P_k already follow independently from Theorem 4.7.

Filter soundness then identifies semantic typing with fixedness: for a semantic type a, a value x has type a precisely when $p(a, x) = x$. In particular $p(a, x) \le x$ is in the image of this projection. We use the usual filter interpretation $[\![M]\!]\rho$, identifying consistent regular filters with their reflected domain elements whenever semantic typing supplies this identification. The following compact-cover property imposes no prescribed stage bound.

Lemma 5.7 (Typed compact cover)**.** *If a compact basis element k satisfies $k \le p(a,x)$, then there merely exist proper compact representatives v, c such that*

$$k \le v \le x, \qquad c \le a, \qquad v : c.$$

Proof. The typed generators are closed under compatible finite joins and compact equivalence, and include bottom. For a join, join the classifiers as well; presupposition and conversion put both values at the common classifier, and compatible-join typing gives the resulting generator. Thus the generators below x form a directed family with supremum $p(a,x)$. Compactness of k gives a generator above it. Replacing its value and classifier by equivalent proper representatives preserves the inequalities and typing. No bounded-stage interpolation is involved. □

Lemma 5.8 (Typed relation laws)**.** *The following implications hold for all semantic a, x and syntactic indices:*

$$\begin{aligned}
\mathsf{Ty}(p(\mathbf{u},a);T,T') &\Longrightarrow \mathsf{Ty}(p(\mathbf{u},a);T',T),\\
\mathsf{Ty}(p(\mathbf{u},a);T,V) \ \wedge\ \mathsf{Ty}(p(\mathbf{u},a);V,W) &\Longrightarrow \mathsf{Ty}(p(\mathbf{u},a);T,W),\\
\mathsf{Elem}(p(a,x);T,M,N) &\Longrightarrow \mathsf{Elem}(p(a,x);T,N,M),\\
\mathsf{Elem}(p(a,x);T,M,N) \ \wedge\ \mathsf{Elem}(p(a,x);T,N,O) &\Longrightarrow \mathsf{Elem}(p(a,x);T,M,O),\\
\mathsf{Ty}(p(\mathbf{u},a);T,T') \ \wedge\ \mathsf{Elem}(p(a,x);T,M,N) &\Longrightarrow \mathsf{Elem}(p(a,x);T',M,N).
\end{aligned}$$

The relation also has head-reduction transport, expansion in both term arguments, and left diagonal closure.

Proof. Prove the five typed implications together at finite relation depth, with the type implications before the term implications at each successor. The semantic projector is eliminated into lower admissible predicates using its universe, product and function cases. In the function case its action has the form

$$p(\pi(a,\varphi), f)\cdot z = p\big(\varphi\cdot p(a,z),\ f\cdot p(a,z)\big).$$

Restrict the tested argument to $p(a,z)$. Idempotence identifies the result of testing at z with that of testing at $p(a,z)$, and puts the output in the corresponding codomain projector. The depth induction hypotheses therefore supply the required argument conversions and output symmetry or transitivity. The type clauses supply the head and component evidence needed to align the tests.

Passing to the invariant requires a typed cover, not an assumption that stage projectors preserve typing. Given a compact observation below $p(a,x)$, Lemma 5.7 supplies a compact value v and classifier c with

$$k \le v \le x, \qquad c \le a, \qquad v : c.$$

By Proposition 4.1, the compact values v and c are each fixed at some stage. Take the maximum of those two stages. The value is fixed by $p(c,-)$, and the classifier is universe-typed. Recovery transfers the premises to the finite-depth relation at this stage; the finite-depth laws apply; recovery transfers the conclusion back. Lowering to k and compact reconstruction prove the laws on the whole projected element. Type symmetry and transitivity use the same argument with the universe projector.

Head transport and term expansion follow from the clauses and head determinism. Left diagonal closure is proved first at finite depth and then passed to the limit. It does not require semantic typing. Semantic soundness transfers the five projected laws to typed filter denotations. □

5.3.1 Fitting environments and related substitutions

An environment fits the empty context automatically. It fits $\Gamma, z : A$ if its restriction ρ fits Γ, the denotation $[\![A]\!]\rho$ is universe-typed, and its last value is semantically typed by $[\![A]\!]\rho$. We write $\rho \models \Gamma$.

Let $\sigma, \sigma' : \Gamma \to \Delta$ be typed substitutions. Relative to a fitting environment ρ, say they are *related* if for each variable z of Γ, with its type A_z expressed in Γ, we have derivations

$$\Delta \vdash \sigma z = \sigma' z : A_z\sigma, \qquad \Delta \vdash \sigma' z = \sigma z : A_z\sigma',$$

and both relations

$$\mathsf{Elem}(\rho z; A_z\sigma, \sigma z, \sigma' z), \qquad \mathsf{Elem}(\rho z; A_z\sigma', \sigma z, \sigma' z). \tag{11}$$

Equivalently, every proper compact observation of each raw environment filter satisfies these relations. Typed symmetry allows reversal of related substitutions; diagonal closure allows replacing a related pair by its left diagonal. These statements hold in a well-formed source context and use only the variable typings supplied by fittingness.

Lemma 5.9 (Restriction to a fitting extension)**.** *Suppose $\rho \models \Gamma$, the substitutions σ, σ' are related, and $\Gamma \vdash A : U$. Suppose the domain induction hypothesis provides*

$$\mathsf{Ty}(a; A\sigma, A\sigma'), \qquad a = [\![A]\!]\rho.$$

Given an arbitrary argument test

$$\Delta \vdash N = N' : A\sigma, \qquad \mathsf{Elem}(x; A\sigma, N, N'),$$

put $y = p(a, x)$. Then $\rho[z \mapsto y] \models \Gamma, z : A$, and $(\sigma, z \mapsto N)$ and $(\sigma', z \mapsto N')$ are related for that extended environment.

Proof. The raw semantic projection representing y is typed by a, since a is universe-typed. Thus the extended environment fits. Since $y \leq x$, downward closure gives the argument relation at $A\sigma$. The conversion law of Lemma 5.8, together with the type relation at a, gives it at $A\sigma'$. More explicitly, the type premise lowers to $p(\mathbf{u}, a)$, and idempotence identifies $p(a, y)$ with y.

Syntactic cross-substitution applied to $A : U$ gives $\Delta \vdash A\sigma = A\sigma' : U$. Symmetry and conversion of the argument equality give the required equality in the other direction at $A\sigma'$. The old variables retain their relations, since substitution in a weakened type ignores the newly substituted argument. These are exactly the conditions for the two extended substitutions. □

5.3.2 Why restricting the argument suffices

For a body M over $\Gamma, z : A$, let $\Lambda_{A,M,\rho}$ denote the semantic binder filter: its finite graph entries have a compact input u typed by $[\![A]\!]\rho$ and an output observed in $[\![M]\!](\rho[z \mapsto u])$. This is the semantic abstraction used both for lambda bodies and for product families.

Lemma 5.10 (Evaluation through the restriction)**.** *Let $a = [\![A]\!]\rho$ and $y = p(a, x)$. Every observation of $\Lambda_{A,M,\rho} \cdot x$ is an observation of $[\![M]\!](\rho[z \mapsto y])$. Hence, whenever the latter is a typed denotation,*

$$\Lambda_{A,M,\rho} \cdot x \leq [\![M]\!](\rho[z \mapsto y]). \tag{12}$$

The same implication holds for any function component below the binder.

Proof. For a compact input presentation u, consider the auxiliary family

$$H(u) = [\![M]\!](\rho[z \mapsto p(a, u)]).$$

It is monotone in the information order. At a graph entry whose input u is typed by a, the principal filter generated by u is fixed by $p(a, -)$. Thus that entry's output belongs to $H(u)$.

An application observation selects finitely many entries of a binder graph under an input observed in x. Induction on the selection, using monotonicity for a single entry and filter intersection for a combined selection, puts the selected output in $H(u_0)$ at that observed input u_0. Since $u_0 \leq x$, projector monotonicity gives $p(a, u_0) \leq p(a, x)$, so the output belongs to the right-hand side of (12). Upward closure in token order handles the final observation. The argument is on raw filter witnesses; after reflection, truncated witnesses are eliminated only into the desired proposition. Compact reconstruction then transfers any lower admissible predicate of the body denotation to the application result. □

There is a second point in the function and family clauses: the induction hypothesis must supply three comparisons, all at the same left type. Use the fitting extension from Lemma 5.9 and write

$$V = B(\sigma, N), \quad Y = [\![M]\!](\rho[z \mapsto y]),$$
$$P = M(\sigma, N), \quad P' = M(\sigma, N'), \quad Q = M'(\sigma', N), \quad Q' = M'(\sigma', N').$$

Here (σ, N) extends σ by N for the new variable. A self induction hypothesis for M at the diagonal base substitution yields $\mathsf{Elem}(Y; V, P, P')$. The cross induction hypothesis for M, M' yields

$$\mathsf{Elem}(Y; V, P, Q'), \qquad \mathsf{Elem}(Y; V, P, Q),$$

the second by the diagonal argument test (N, N). Typed symmetry and transitivity on Y now give

$$\mathsf{Elem}(Y; V, P, P'), \qquad \mathsf{Elem}(Y; V, Q, Q'), \qquad \mathsf{Elem}(Y; V, P, Q'). \tag{13}$$

This obtains the right self comparison without changing its classifier to $B(\sigma', N)$ and without choosing a uniform family of witnesses.

5.4 The fundamental lemma and product inversion

Theorem 5.11 (Fundamental lemma)**.** *Suppose Δ is well formed, $\rho \models \Gamma$, and $\sigma, \sigma' : \Gamma \to \Delta$ are related relative to ρ. For every derivation $\Gamma \vdash M = N : A$,*

$$\begin{aligned} &\mathsf{Elem}([\![M]\!]\rho; A\sigma, M\sigma, N\sigma'), \\ &\mathsf{Elem}([\![M]\!]\rho; A\sigma', M\sigma, N\sigma'). \end{aligned} \tag{14}$$

For a typing derivation $\Gamma \vdash M : A$, the same conclusions hold with $N = M$. They hold on the whole reflected denotation and on every compact observation below it.

Proof. Induct simultaneously on typing and equality derivations, with the well-formed-context cases contributing no relational assertion. The hypotheses are quantified over all fitting environments and related substitutions. We give the cases that explain the construction.

Variables, universe, and structural rules. The variable case is (11). The universe case uses its own head reduction and the absence of a product reduct of U. Reflexivity uses the typing hypothesis. Symmetry reverses the related substitutions, applies the induction hypothesis, and uses typed symmetry; semantic soundness identifies the denotations of the equal terms. Transitivity uses the diagonal substitutions and typed transitivity. Conversion applies the type induction hypothesis and the conversion law in Lemma 5.8, at each substituted classifier.

Dependent products. For formation, the premises are $\Gamma \vdash A : U$ and $\Gamma, z : A \vdash B : U$. For equality they additionally include $\Gamma \vdash A = A' : U$ and $\Gamma, z : A \vdash B = B' : U$, with the specified typings of the bodies. Substitution and cross-substitution give the syntactic component equalities

$$\Delta \vdash A\sigma = A'\sigma' : U, \qquad \Delta, z : A\sigma \vdash B(\sigma^+) = B'(\sigma'^+) : U.$$

Here σ^+ denotes the lifted substitution under the binder. These equalities come from the premises, not from inversion of a product equality. The domain induction hypothesis gives the type relation on $[\![A]\!]\rho$.

The product denotation has no universe head. Under a positive product observation, its extracted domain is below $[\![A]\!]\rho$, and its extracted family is below $\Lambda_{A,B,\rho}$. For an arbitrary related argument, Lemma 5.9 gives a fitting extension. The body induction hypotheses and (13), with classifier U, supply the three type comparisons in (6). Lemma 5.10 transfers them to the extracted family. Fold the type clause, then the term clause at U. Formation is the self-equality instance. Since the classifier is U, the two substituted classifier conclusions coincide.

Annotated abstraction. For $\lambda A.M : \Pi(A, B)$, its semantic value has only a function component, so the type component of the term relation has no positive head to test. For a related argument, restrict it as above, use the body induction hypothesis and (13), and transfer the outputs by (12). Syntactic beta reduction and term expansion replace the instantiated bodies by applications of the substituted abstractions. This proves the three function outputs at $\Pi(A, B)\sigma$. Apply the same argument with σ, σ' reversed and then typed symmetry on the abstraction denotation to obtain the conclusion at $\Pi(A, B)\sigma'$.

Application and beta. The function relation applied to the argument relation yields the application comparison at the left substituted result type. Instantiating the codomain induction hypothesis under the binder gives the relation between the two substituted result types; typed conversion gives the right-type conclusion. This treats ordinary application and both congruence rules. For beta, extend the environment by the argument denotation and the substitutions by the two substituted arguments. The body hypothesis applies there. Semantic beta and substitution identify the denotation, while syntactic beta and term expansion give the required equality comparison at both result types.

Typed function extensionality. Suppose the premises include

$$\Gamma, z : A \vdash f^\uparrow z = g^\uparrow z : B, \qquad \Gamma \vdash f : \Pi(A, B), \quad \Gamma \vdash g : \Pi(A, B).$$

Semantic typing gives the eta inclusions

$$[\![f]\!]\rho \leq \Lambda_{A,f^\uparrow z,\rho}, \qquad [\![g]\!]\rho \leq \Lambda_{A,g^\uparrow z,\rho}. \tag{15}$$

To see the first, take a typed compact cover of an observation of $[\![f]\!]\rho$. A non-bottom cover must be a function graph: universe and product values cannot have a product classifier. Each graph entry has a key typed by $[\![A]\!]\rho$ and its output is observed in the application of $[\![f]\!]\rho$ to that key. Semantic weakening identifies this with the corresponding eta body. All entries therefore belong to the binder, and the graph covers the original observation. The proof for g is the same. This is a consequence of semantic typing and soundness.

The pointwise equality induction hypothesis, together with its symmetry and transitivity consequences, supplies the self and cross hypotheses for the two eta bodies. Restrict each tested argument to a fitting extension, obtain (13), and use (15) and Lemma 5.10 to transfer them to applications of $[\![f]\!]\rho$. Substitution in $f^\uparrow z$ is exactly application of $f\sigma$ to the substituted argument. Hence the three outputs fold to the required function relation. For the right

classifier, reverse both the pointwise equality and the substitutions, use typed symmetry, and transfer from the denotation of g to that of f by semantic soundness of the extensionality rule.

These cases cover every derivation constructor. The proof can be read observation by observation on the raw filters. Semantic typing gives consistent regular denotations, and Lemma 5.2 then supplies the whole-denotation assertions. Conversely, downward closure gives every compact-observation conclusion. □

Corollary 5.12 (Mere product inversion)**.** *From a derivation* $\Gamma \vdash \Pi(A, B) = \Pi(A', B') : T$*, for any classifier* T*, one obtains*

$$\left\|(\Gamma \vdash A = A' : U) \times (\Gamma, z : A \vdash B = B' : U)\right\|.$$

Proof. The environment $\rho_\perp$ assigning bottom to every variable fits every well-formed context. Prove this by induction on the context: soundness makes the next type universe-typed, and bottom is a value of that type. The identity substitution is related to itself at $\rho_\perp$ by strictness of the relation.

The literal product denotation has the compact observation $k_\Pi = \pi(\perp, \perp)$: use the bottom observation of its domain and the empty family graph. Apply Theorem 5.11 at the identity substitutions and observe k_Π. The type component of the term clause gives $\mathsf{Ty}(k_\Pi; \Pi(A, B), \Pi(A', B'))$. Unfolding the positive product clause supplies a mere tuple of evidenced heads with their component equalities. Literal products are head normal, so head determinism identifies those components with A, B, A', B'. Mapping this truncated tuple to its equalities gives the displayed conclusion. □

The argument derives product inversion after the fundamental lemma; it uses neither syntactic product injectivity nor an earlier adequacy theorem as a premise. The conclusion is explicitly truncated. It does not assert a chosen pair of derivation trees, or a simultaneous choice of witnesses over all compact observations.

A Embedding–projection categories and initial algebras

In this appendix we also use an *unpresented* Scott domain: its small compact basis is merely known to exist, rather than selected as object data. A presentation supplies such an object by propositional truncation.

Proposition A.1 (Univalence of Scott-domain objects)**.** *Let D and E be unpresented Scott domains. The canonical map is an equivalence*

$$(D = E) \simeq \{\textit{order isomorphisms between } D \textit{ and } E\}.$$

Proof. Suprema are unique, so completeness contributes no choices for a fixed order. Least elements are unique, and the basis-existence condition is a proposition. It therefore suffices to compare orders. Fix an order P. The total space of orders equipped with an order isomorphism from P is contractible: univalence makes the type of carrier equivalences from P contractible, while propositional univalence and the order laws make the relation over each such equivalence unique. The total space for Scott domains is a retract of this space equipped with the remaining structure in proposition-valued fibres; it is contractible as well. Projecting its unique path from (D, id) to (E, i) produces the inverse of the canonical map. Path induction and contractibility give both inverse laws. □

Proposition A.2 (Univalent categories of Scott domains)**.** *Unpresented Scott domains form a univalent category with Scott-continuous maps. They also form a univalent category whose morphisms $D \to E$ are embedding–projection pairs of continuous maps $e : D \to E$ and $p : E \to D$ satisfying $p \circ e = \mathrm{id}_D$ and $e \circ p \leq \mathrm{id}_E$ pointwise. In both categories, categorical isomorphisms correspond to order isomorphisms of domains.*

Proof. Identity and composition preserve continuity. For pairs, composition is $(e_2 \circ e_1, p_1 \circ p_2)$; monotonicity and the two pair laws give its retraction and pointwise inequality. An invertible continuous map has a continuous inverse: an order isomorphism preserves and reflects directed suprema. Conversely, the underlying map of a categorical isomorphism and its inverse give an order isomorphism. For an invertible pair, its projection agrees with the inverse of its embedding, so its entire arrow is determined by that order isomorphism. Thus the singleton total space of categorical isomorphisms from a fixed domain is a retract of the proposition-valued total space used in the preceding result. The total-space criterion for univalence applies to both precategories. □

Theorem A.3 (Compact elements and ideal completion)**.** *Let $\mathbf{Dom}_{\lhd}$ be the univalent category of unpresented Scott domains and continuous embedding–projection pairs. Let* **CUSL** *be the univalent category of conditional upper semilattices with merely small carrier. Its morphisms preserve bottom and compatible binary joins and reflect both order and compatibility. Then*

$$\mathbf{Dom}_{\lhd} \simeq \mathbf{CUSL}.$$

The functors are $\mathsf{K}(D)$, the CUSL of compact elements, and $\mathrm{Idl}(P)$, the Scott domain of ideals of P. There are natural isomorphisms

$$P \cong \mathsf{K}(\mathrm{Idl}(P)), \qquad D \cong \mathrm{Idl}(\mathsf{K}(D)).$$

Proof outline. For a small presentation of P, ideals are inhabited, downward closed, directed lower subsets. Inclusion is a small relation. Directed suprema are unions; the supremum of a bounded family is the ideal generated by its finite joins. Principal ideals are compact and form a small basis. An embedding of CUSLs extends to the downward closure of its direct

image; inverse image supplies the continuous projection. The four embedding laws give the retraction and deflation equations.

Conversely, an embedding–projection pair sends compact elements to compact elements: its embedding is left adjoint to its projection, and the projection preserves directed suprema. The restricted embedding preserves compatible joins and reflects compatibility. A small basis presents the compact-element carrier via the replacement image; the presentation is only asserted to exist, so it does not become object data. For a set, univalence makes two small presentations equal. Thus mere smallness supplies the small presentation used for ideal completion without a choice axiom. Every compact ideal is principal, while each domain element is the supremum of the compact elements below it. These give the displayed isomorphisms. Their naturality is checked on principal ideals and on the projections of embedding–projection pairs, respectively. □

A.1 Countable chains of presented domains

Proposition A.4 (Countable limits of presented embedding–projection chains)**.** *Let $D_0 \to D_1 \to \cdots$ be a chain of presented Scott domains and continuous embedding–projection pairs (e_n, p_n). The coherent sequences*

$$L = \{(x_n)_n \mid p_n(x_{n+1}) = x_n\}$$

form a presented Scott domain. Each D_n embeds into L, with projection given by the nth coordinate, and every $x \in L$ is the directed supremum of its stage approximants $e_{n,\infty}(x_n)$. The compact basis consists of stage lifts of compact basis elements. This cone is a colimit in the embedding–projection category on presented objects: every compatible cone to a presented Scott domain has a unique mediating pair.

Proof. Orders and directed suprema of coherent sequences are coordinatewise. For bounded suprema, take the supremum at each stage, lift it to L, and then take the directed supremum of these stage lifts. A stage lift is compact when its stage element is compact. Stage approximation and compact density at each stage give density in L without countable choice.

For a compatible cone $(u_n, v_n) : D_n \to Y$, the mediating projection sends y to $(v_n(y))_n$; its embedding sends $(x_n)_n$ to $\bigvee_n u_n(x_n)$. The cone equations make the latter family increasing. The retraction and deflation laws follow by checking each stage coordinate and using continuity. A second mediating pair agrees on each stage lift, so stage exhaustion and continuity make its embedding equal to this one; its projection is equal coordinatewise.

□

A.2 Initial algebras and categorical transfer

Theorem A.5 (Abstract initial-chain theorem)**.** *Let $\mathcal{C}$ be a category with an initial object and colimits of countable chains. If an endofunctor $F : \mathcal{C} \to \mathcal{C}$ preserves these colimits, then the colimit μF of*

$$0 \longrightarrow F0 \longrightarrow F^2 0 \longrightarrow \cdots$$

has an initial F-algebra structure $\mathsf{fold} : F(\mu F) \to \mu F$. The map fold is an isomorphism. If $\mathcal{C}$ is univalent, the isomorphism induces an equality $\mu F = F(\mu F)$.

Proof. Removing the initial object from the chain leaves a chain with the same colimit: a cocone over the tail extends uniquely at stage zero. Preservation by F makes $F(\mu F)$ another colimit of this tail. The unique mediating arrows between these two colimits give fold and unfold; their composites are identities by colimit uniqueness.

For an algebra $a : F(Y) \to Y$, define its stage maps recursively by the unique arrow $0 \to Y$ and $h_{n+1} = a \circ F(h_n)$. The functor laws make them a cocone. Its mediator $h : \mu F \to Y$ satisfies $h \circ \mathsf{fold} = a \circ F(h)$ because both sides agree on every image stage and $F(\mu F)$ is their colimit. Any other algebra morphism agrees with the stage maps by induction, hence equals h by colimit uniqueness. Since hom types are sets, the algebra equation is a proposition, so the type of algebra morphisms is contractible. □

The theorem is formalized for precategories, with a separate generic univalent-category corollary for the object equality.

Theorem A.6 (Countable CUSL colimits and initial algebras)**.** *The univalent category* **CUSL** *of conditional upper semilattices and embeddings has an initial object and colimits of countable chains. For any endofunctor $F : \mathbf{CUSL} \to \mathbf{CUSL}$ preserving these colimits, there is an initial F-algebra $\mathsf{fold} : F(\mu F) \to \mu F$. Its structure map is an isomorphism, and univalence gives $\mu F = F(\mu F)$. The preservation hypothesis is a proposition.*

Proof. The one-point CUSL is initial. For a chain of CUSL embeddings, form the chain of their presented ideal domains and its coherent-sequence EP limit. Take the compact elements of this limit as the colimit CUSL. The unit of the CUSL–EP equivalence and the canonical EP cone give the cocone legs. For a cocone into Y, ideal extension gives an EP cocone into the presented ideal domain of Y; its universal mediator restricts to a CUSL embedding from the compact elements of the limit to Y. Naturality of the unit gives the factorization equations. Every compact element of the limit is equal to the lift of a compact basis element at some stage, so agreement on the cocone legs determines the mediator uniquely. The abstract initial-chain theorem now supplies the initial algebra and the equality.

For a fixed cocone, two mediators satisfying the universal property agree by uniqueness. All remaining laws are equalities in hom sets, so its colimit structure is a proposition. Function extensionality then shows that preservation of all existing countable-chain colimits is a proposition. The small presentations used in ideal completion are selected by univalence from mere smallness; no countable choice is used. □

Theorem A.7 (Categorical structure identity principle)**.** *At the object and hom universe levels used here, an equivalence between univalent categories $\mathcal{C}$ and $\mathcal{D}$ induces an equality $\mathcal{C} = \mathcal{D}$ of category records. In particular,* $\mathbf{CUSL} = \mathbf{Dom}_{\triangleleft}$.

Proof. This is the equivalence-to-identity direction of the categorical structure identity principle of Ahrens–Kapulkin–Shulman. The two natural reconstruction isomorphisms give mutually inverse functions on object types after category univalence converts their components to object equalities. Univalence of the object universe then identifies the object types. The functor induces an equivalence on each hom type: its inverse is obtained by applying the quasi-inverse functor and conjugating by the reconstruction isomorphisms. Fiberwise univalence identifies the hom families. The functor laws identify identities and composition after transport. Because hom types are sets, the remaining category laws are propositions; univalence of a fixed precategory is also a proposition. The resulting equality of precategories extends to equality of category records. The checked construction is `CategorySIPTheorem.On.equality`; it uses only the previously stated universe univalence and function extensionality. □

Corollary A.8 (Unpresented Scott EP initial algebras)**.** *The category of unpresented Scott domains and embedding–projection pairs has colimits of countable chains. Every endofunctor of this category that preserves these colimits has an initial algebra with an invertible structure map and an object equality $X = F(X)$.*

Proof using univalence. By Theorem A.7, the CUSL–EP equivalence gives an equality of categories. Transport the CUSL countable-chain colimit structure along this equality; this is the definition `ScottEPTransport.cocomplete`. The endofunctor F remains an endofunctor of EP throughout: only the colimit structure is transported. The one-point EP domain is initial, so Theorem A.5 applies. Univalence turns its solution isomorphism into equality of objects. □

Proof up to isomorphism using a precategory equivalence. The categorical argument needs only an equivalence of precategories $I : \mathbf{CUSL} \rightleftarrows \mathbf{Dom}_{\lhd} : K$ with natural reconstruction isomorphisms, together with the CUSL chain colimits. It does not need an equality of the two category records. For an EP chain S, take a colimit (L, λ_n) of the chain KS. The EP colimit vertex is IL, with legs $I(\lambda_n) \circ \eta_{S_n} : S_n \to IL$, where $\eta_{S_n} : S_n \cong IK(S_n)$ is the reconstruction isomorphism. Given a cocone $q_n : S_n \to Y$, the arrows $K(q_n)$ have a unique mediator $m : L \to KY$. Compose $I(m)$ with the inverse of $\eta_Y : Y \cong IKY$ to obtain a mediator $IL \to Y$. Naturality of η proves the factorization equations; full faithfulness of I proves uniqueness. Thus EP has countable-chain colimits as a precategory. Apply Theorem A.5 to the initial EP object and an endofunctor F preserving these colimits. Its initial-chain colimit X has inverse EP arrows $\mathsf{fold} : F(X) \to X$ and $\mathsf{unfold} : X \to F(X)$, so $X \cong F(X)$. Here continuity of F means preservation of countable-chain colimits. Function and propositional extensionality suffice for the equalities of maps and proof fields in this argument. If objects are passed to an isomorphism-class quotient, it also gives $[X] = [F(X)]$ there; an equality $X = F(X)$ of domain objects uses the univalence step in the preceding proof. □

Remark A.9 (Scope of the formalization without univalence). The precategory transfer is checked abstractly in `CategoryPreEquivalenceColimits`; the initial-chain construction is checked for arbitrary precategories in `CategoryInitialChain`.

The current concrete construction of I and of the CUSL colimits is inside the development parameterized by univalence. In particular, `ScottCUSLIdeals.Canonical` calls `ScottSmallness.choose` to extract a small presentation from mere smallness; that operation uses univalence of the small universe. The checked concrete transfer to presented EP also uses this canonical completion. Consequently, the abstract second proof is formalized, but the full concrete corollary for unpresented Scott EP has not been separately checked from only propositional extensionality, function extensionality, and quotients. A construction of the needed ideal completion from those assumptions alone remains to be supplied.

Corollary A.10 (Transport of a CUSL domain equation)**.** *Let $F : \mathbf{CUSL} \to \mathbf{CUSL}$ preserve countable-chain colimits. Under the equality $\mathbf{CUSL} = \mathbf{Dom}_{\lhd}$ of Theorem A.7, transport of F and its initial algebra gives an EP endofunctor F^{EP} with an initial algebra $\mathsf{fold}^{\mathrm{EP}} : F^{\mathrm{EP}}(\mu^{\mathrm{EP}} F) \to \mu^{\mathrm{EP}} F$. Its fold map is invertible and $\mu^{\mathrm{EP}} F = F^{\mathrm{EP}}(\mu^{\mathrm{EP}} F)$.*

Proof. Theorem A.6 gives a dependent package comprising F, its colimit-preservation proof, the initial F-algebra, invertibility of its structure map, and the object equality. Transport this whole package along the category equality. The resulting package has exactly the stated fields for an EP endofunctor. The checked construction is `ScottCUSLSolutionTransport.On`; its algebra-morphism center and uniqueness fields establish initiality after transport. □

Theorem A.11 (Univalence of the algebra category)**.** *For any univalent category $\mathcal{C}$ and endofunctor $F : \mathcal{C} \to \mathcal{C}$, the category $\mathrm{Alg}(F)$ is univalent. Its objects are pairs $(X, a : F(X) \to X)$, and its arrows $h : (X, a) \to (Y, b)$ satisfy $h \circ a = b \circ F(h)$.*

Proof. An algebra isomorphism gives an isomorphism of underlying objects, hence an equality $X = Y$ by univalence of $\mathcal{C}$. The commuting square identifies the transported algebra map

with b, giving equality of the algebras. Path induction and the sethood of hom types show that this construction and the canonical map from equalities to isomorphisms are inverse. The functor laws give identities and composition of algebra arrows. No colimit preservation assumption on F is used. □

The direct coherent-sequence construction uses *presented* Scott domains. Transport by category equality extends countable-chain colimits to arbitrary unpresented objects with merely existing compact bases. Here "ω-continuous" means preservation of existing countable-chain colimits. An implication from continuity on ordered hom types to this preservation is not asserted. The preservation property is a proposition. A separate generic proof transports countable-chain colimits directly along a categorical equivalence, without using equality of categories; the corollary above uses the stronger structure identity theorem.

Formal checks and total timing

The development is checked in Unit with univalence and replacement as ambient parameters. Function extensionality is derived from univalence; propositional truncation is obtained from the replacement interface. No excluded middle, choice or propositional resizing axiom is added. The results discussed here are drawn from the following checked developments:

1. Scott domains with small compact bases, including a univalent object type with merely existing basis and univalent categories of continuous maps and embedding–projection pairs; continuous function spaces, compact step approximations, least fixed points, and the abstract initial-chain theorem, Theorem A.5, and the generic univalent category of endofunctor algebras; the equivalence of univalent Scott EP and CUSL categories, Theorem A.3. The CUSL initial-algebra theorem, Theorem A.6, uses the checked countable-chain colimit construction; the categorical structure identity principle, Theorem A.7, transfers it to the unpresented EP corollary, Corollary A.8, and directly transports each CUSL solution by Corollary A.10.

2. The Scott filter domain and recursive projector, with independent preservation of normalized finitary families and their Kleene limit (Theorem 4.7); the finite-stage projectors and their structural laws, finite stabilisation, and comparison with bounded compact typing. The existing Unit proof constructs these projectors using ranks of finite presentations. It also checks the recursive successor equation used in Proposition 4.1. A proof of all the tower laws starting solely from that recursive definition has not been separately formalised.

3. The independent strict lower admissible invariant, its equations and uniqueness; typed symmetry, transitivity and conversion.

4. Typed argument restriction; Pi formation and equality, annotated lambda and function extensionality; the complete derivation induction.

5. Whole-denotation adequacy at both substituted types, compact recovery, and mere syntactic Pi inversion.

The complete induction is `At.fundamental` in `IndependentFundamental.seg`; its consequences are in `IndependentAdequacy.seg` and `IndependentInversion.seg`.

On 13 November 2026 the updated `python3 UNIT/check.py` passed the full development (14,856 declarations), the general dependent-product and dependent-sum session (1,156), the independent projector session (3,838), and the independent logical-relation session (6,338), each with zero failures. The general dependent-product session has no filter-domain or typing modules in scope. The projector session excludes the semantic projector, typed stage comparisons, and the older direct common-code proof. Both import boundaries are audited by the script. All 40 dependency hashes remain unchanged. The previously recorded timing below concerns the older two-session check.

On 14 November 2026, after adding admissible fixed-point induction and using it for the projector invariant, the same script passed all four sessions: 14,883, 1,156, 3,849, and 6,349 declarations respectively, each with zero failures. All 40 frozen dependency hashes matched.

After the subsequent general CUSL construction of Fp on 14 November, the script passed five sessions: the full development (15,724 declarations), the independent CUSL Fp session (2,412), the general product/sum session (1,156), the independent projector session (3,849), and the independent logical-relation session (6,349), each with zero failures. The CUSL session checks the general functor, small directed-colimit preservation, adjunction and action formulas with the filter and typing modules out of scope. Its import boundary is audited by the script. All 40 dependency hashes matched.

All five sessions also pass with a checker built from the published Unit revision `18bb219f`. Reproduction instructions and the full revision identifier are in `UNIT/README.txt`.

A run on 12 November 2026 of `/usr/bin/time -p python3 UNIT/check.py`, using the recorded local macOS arm64 checker, gave the following results. The type-checker is a compiled Haskell program. Python only verifies the dependency hashes and invokes that program for the two checks; `time -p` measures the complete command.

Check	Declarations	Checker CPU time
Complete development	12,187	8.56 s
Independent relation and consequences	4,909	1.18 s

The **total wall-clock time was 10.46 seconds**, including dependency verification and both checker processes. The independent run has a 104-module dependency closure with the old logical relation and the earlier proofs of adequacy, syntactic Pi injectivity and subject reduction out of scope. All 40 frozen dependency hashes were verified unchanged. The two declaration counts overlap; the full count also includes developments outside this account. These timings describe this local run, not a machine-independent bound.

For comparison, on 14 November 2026 we also checked the original MIN proof of domain-model soundness, logical-relation adequacy, Π-injectivity and subject reduction. Full-source checks took **0.54 seconds wall-clock time in Unit** (1,800 declarations, zero failures; 0.38 seconds checker CPU time) and **26.42 seconds in Agda 2.8.0** (54 modules). The Agda command used `-ignore-interfaces -without-K` on `MIN/SubjectReduction.agda`; Unit checked the original `domain/mltt/All.seg`, including both versions of its logical relation. These sequential runs concern the two implementations of the earlier finite-element proof; the larger Scott-domain development timed above also contains the additional constructions described in this note. Commands and measurements are recorded in `UNIT/MIN_TIMINGS.txt`.

The inversion conclusion remains propositionally truncated. Comparison with the old uniform proof-relevant relation and transfer to the stable reflexive domain remain separate. The checked projector tower has not been identified with the separately constructed CUSL initial-chain projections; that comparison requires finite-stage coverage and compatibility with application. The classical finite-poset counterexample is illustrative and has not been formally checked.

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